The phenomenology of hallucinogenic experiences: a mixed-methods comparison of psilocybin and brugmansia

with a two-stage scene-individuation and attribute-coding framework

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Abstract

We present a phenomenological comparison of psilocybin and brugmansia hallucinatory experiences derived from human-coded trip reports. Each of 40 trip reports (20 psilocybin, 20 brugmansia) was independently coded by two hypothesis-blind raters against a 147-item hierarchical taxonomy of perceptual, cognitive, emotional, bodily, and existential features. A post-hoc consistency pipeline then reorganised the raw codings into a scene-ID-based dataset that makes inter-rater agreement separately measurable at two tiers: (i) scene individuation — did both raters see the same passage as a hallucinatory scene — and (ii) attribute classification — did they tag it with the same taxonomy items. A conservative consensus filter, retaining only scenes both raters individuated and items both raters attached, operationalises the original coding instructions' Case-2 conservative rule at the dataset level and yields a high-confidence subset for between-substance comparison.

1 Introduction

1.1 Background

1.2 Hypotheses

2 Methods

This section describes (i) the source corpus of trip reports, (ii) the original two-phase human coding procedure, and (iii) the post-hoc LLM-assisted consistency pipeline that converted the raw rater spreadsheets into a scene-identifier-based dataset suitable for quantitative analysis.

2.1 Corpus and coders

The corpus comprises 40 first-person trip reports of two hallucinogenic substances: 20 reports of psilocybin (*Psilocybe* spp.) and 20 reports of brugmansia (*Brugmansia* spp.). Each block of 20 reports was divided into two sub-blocks of 10 reports, and each sub-block was independently coded by two hypothesis-blind raters drawn from a pool of eight coders (Table 1). Raters were unaware of the study hypotheses, received identical written guidelines, and had no contact with one another during coding.

Trip reports were sourced from public psychonaut archives and de-identified. Each report carries a unique substance-specific identifier (e.g. `Psilocybe_01`), an optional self-reported dose, and a narrative of variable length (349–5 531 words; median ≈1 500).

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Substance & block	Coder A	Coder B
Psilocybin 01–10	Alessandra	Brendan
Psilocybin 11–20	Francesco	Susana
Brugmansia 01–10	Brendan	Noah
Brugmansia 11–20	Alessandra	Alessio

Table 1: Coder-pair assignments across the four blocks.

2.2 Coding system

Raters coded each report against a taxonomy of 147 items organised hierarchically into five high-level sections and 23 mid-level categories:

- **Perceptual changes**: type of visual alteration (illusion, incrusted object, detached object, full-blown immersive), visual/auditory/tactile/olfactory/gustatory/nonsensory hallucinations (each sub-classified by object class: human, artefact, animal, plant, inorganic entity, extraordinary entity), modal status (possible/probable, possible/improbable, impossible), dynamics (object constancy vs no object constancy).
- **Sensory changes**: sensory delight (visual, auditory, haptic, erotic), synaesthesia, enhanced mental imagery.
- **Cognitive and emotional changes**: sense of reality (lucid, delusional), amnesia, domain-specific violations (biological and psychological projection), emotion (positive, negative).
- **Bodily and autonomic changes**: arousal (abnormally high/low), out-of-body experience, own body distortion.
- **Existential changes**: insight (metaphysical, religious, practical, existential), ego dissolution (social, bodily, core self), sense of surprise and unexpectedness, sense of strangeness and enigma.

A subset of the taxonomy is *scene-level* (hallucination modalities, alteration type, modal status, dynamics): each of these items is attached to a specific hallucinatory episode within the trip. The remainder is *trip-level* (emotion, arousal, insight, ego dissolution, etc.): these items describe the trip as a whole. The "Judgment of reality" row, present in the original spreadsheet, was marked deprecated in the instructions and was not coded.

2.3 Original human coding procedure

Raters followed the procedure specified in the project's written guidelines. In outline:

Step 1: scene individuation. The atomic unit of coding was the *hallucinatory scene*, defined as a coherent episode in which the narrator perceives something that is not really present (a hallucination) or misperceives a real object's properties (an illusion). Per the guidelines, a scene is not defined by sentence or object boundaries: "a scene can change a little and still remains the same scene". Moving from a hallucinated room to a hallucinated palace with a king on a throne, by contrast, counts as a new scene. Raters parsed each report and enumerated its scenes.

Step 2: attribute attachment. For each scene, the rater attached every taxonomic item that was clearly instantiated. Object-class items were counted once per scene regardless of the number of instances (3 000 hallucinated warriors in one scene are coded as "Human | Unknown person(s)" with count 1). If a hallucination appeared in multiple modalities (e.g. a visual figure who also spoke), each modality was tagged independently.

Step 3: trip-level items. Items describing the trip as a whole (emotion, arousal, insight, ego dissolution, amnesia, domain-specific violations) were attached at trip level without scene attribution.

Conservative rule. The guidelines specified a conservative coding discipline: if it was not clear whether an item was instantiated in the text, raters were to treat it as not instantiated and leave the cell blank. This is described in the guidelines as "Case 2": if unclear, do not code; only code passages that are relatively explicit.

Each rater recorded their codings in a spreadsheet with one row per taxonomy item and, for each item, an integer count of occurrences in column E and verbatim excerpts in subsequent columns.

2.4 Post-hoc LLM-assisted consistency pipeline

The eight rater spreadsheets represent the same 40 trips coded twice each, but their row-by-row structure makes inter-rater disagreement difficult to decompose. A disagreement can arise from (i) the two raters individuating scenes differently, (ii) agreeing on a scene but attaching different taxonomy items to it, or (iii) simply quoting different lengths of the same passage as the exemplifying excerpt. The spreadsheet format conflates these sources of disagreement into a single per-item cell.

To disentangle them we implemented a six-stage consistency pipeline in which a large language model (Claude Opus 4.6; **?**) was used under human direction for the one step that intrinsically requires semantic judgement — matching each rater's excerpts to a canonical set of scenes per trip — while all other stages were deterministic and scripted. All pipeline outputs are stored in the `1.Recoding/` directory of the project repository; the canonical source spreadsheets and trip-report `.docx` files were never modified.

Stage 1: taxonomy extraction. The 147-item taxonomy was parsed from one representative rater spreadsheet (the most complete; `Psilocybe_11-20_Susana.xlsx`) and augmented with a hierarchical annotation (`item_id`, `parent_id`, `depth` $\in \{0, 1, 2\}$, `is_leaf`, `has_children`, and an `is_scene_level` flag). Identical taxonomies were verified across all eight rater sheets (zero structural differences). The result is `data/taxonomy.csv`.

Stage 2: lossless raw extraction. Each of the eight rater spreadsheets was parsed into a long-format CSV (`data/raw_codings.csv`, $n = 2\,395$ rows) with one row per (rater, trip, item, excerpt) tuple. Trip identifiers were normalised to the canonical form `Substance_NN` (e.g. stripping stray hashes, whitespace, and the four blank trip-IDs in one spreadsheet were inferred by position). This CSV is a lossless mirror of the original rater data; no information is discarded.

Stage 3: narrative extraction. The four trip-report `.docx` files were parsed and split by the in-text `ID:` markers into 40 per-trip plain-text narratives, stored in `data/trips/`. Per-trip word counts and doses were extracted into a `trips_index.csv` manifest.

Stage 4: adjudication-worksheet generation. For each trip, a markdown "adjudication worksheet" was auto-generated that displayed (a) the full trip narrative, (b) every scene-level excerpt recorded by rater A with its approximate character-span anchor in the narrative (computed via longest-common-substring and `rapidfuzz` partial-ratio fallback), and (c) the corresponding list for rater B. Excerpts attached by the same rater to multiple taxonomy items were deduplicated to a single row to make the raters' implied scene-list visible at a glance. The 40 worksheets live in `worksheets/`.

Stage 5: per-trip scene adjudication. This is the stage where semantic judgement is required and where automatic matching algorithms are unsuitable: two raters' excerpts must be compared against the underlying narrative to decide whether they refer to the same hallucinatory scene. We used Claude Opus 4.6 in a structured human-in-the-loop protocol, one trip at a time:

1. The worksheet, the source narrative, and the project's coding-instruction PDFs were provided as context.
2. The model produced a YAML adjudication file per trip listing a canonical set of scenes, each with a stable scene identifier (see naming convention below), a short description, the character-span in the narrative, the worksheet-excerpt references contributed by each rater, and the taxonomic items each rater had attached.
3. The human co-author (G.F.) read each YAML against the source narrative and the worksheet and reviewed the model's unification/splitting decisions, accepting or redirecting them. Redirections were fed back into the model to refine the convention and to retrofit already-adjudicated trips where a newly-recognised pattern applied.

A formal naming convention encodes rater status directly in each scene's identifier, making inter-rater structure queryable from the data alone. Every canonical scene has an identifier of the form `{trip_id}_S{NN}[.k]_{STATUS}[_MOD]`:

- **Status suffix**: `_AB` when both raters individuated the scene, `_A` when only rater A individuated it, `_B` when only rater B did.
- **Optional structural modifier**: `_LUMPA` and `_LUMPB` flag a parent scene where one rater lumped what the other split; corresponding child scenes carry a dotted index (`S08.1`, `S08.2`, ...) and the `_SPLITB` / `_SPLITA` modifier.
- A `parent_scene_id` column in the resulting `scenes.csv` links children back to their lump.

This convention makes three distinct inter-rater disagreement tiers separately filterable from the data:

Tier 1 (scene individuation): Did both raters individuate this scene? Answered directly by the `_AB`/`_A`/`_B` suffix.

Tier 2 (attribute classification): Given that both raters individuated a scene (`_AB`), did they attach the same taxonomy items to it?

Tier 0 (binned view, analytical only): Did both raters see *anything* in this narrative region, regardless of how they chunked it? Derived at analysis time by grouping scenes with overlapping character-spans into "bins" (see Stage 6). This view is computed on demand and does not modify the canonical dataset.

The naming convention was applied retroactively: when a new pattern was discovered during adjudication of a later trip, earlier adjudications were revisited to ensure consistent application across the full dataset. The conventions are documented in `RULES.md`, which functions as an iterative living specification.

Stage 6: consolidation and derived tables. A deterministic consolidation script ingests the 40 adjudication YAMLs and produces four canonical long-format CSVs plus two derived tables:

- `trips.csv` (40 rows): one row per trip with substance, block, coder pair, dose, word count.
- `scenes.csv` (305 rows): one row per canonical scene, with `scene_id`, `rater_status`, canonical span, per-rater excerpt references, and any lump/split metadata.

- `codes.csv` (1 546 rows): one row per (scene or trip, rater, item) attachment, joined to taxonomy hierarchy.
- `agreement_flags.csv` (754 rows, derived): for every shared scene, every taxonomy item that at least one rater attached is listed with `A_coded`, `B_coded` booleans and an `agreement` $\in \{\text{AGREE}, \text{A_ONLY}, \text{B_ONLY}\}$ flag.
- `rater_style.csv` (80 rows, derived): per-trip, per-rater counts quantifying style asymmetries (scenes individuated, scene-codes attached, modal-status uses, incrusted/detached/illusion/immersive tag uses, codes-per-scene density).
- `analysis/scene_bins.csv` and `analysis/bins_summary.csv` (derived, non-canonical): the binning analytical view — scenes with overlapping canonical spans grouped into bins, with a bin-level status (both / only_A / only_B) and flags for granularity-recovered agreement and chunking-split patterns. These files live in `analysis/` to signal that they are a view, not a mutation of `scenes.csv`.

2.5 Conservative consensus filter

Quantitative between-substance comparisons in this paper use a conservative consensus filter that operationalises the original guidelines' Case-2 rule at the dataset level. An observation is retained if and only if:

1. `scenes.rater_status == "both"` (both raters individuated the scene), **and**
2. `agreement_flags.agreement == "AGREE"` (both raters attached the item to the scene).

The justification is direct: an item coded by only one rater is, by construction, a Case-2 passage — at least one of the two blind raters did not find the evidence clear enough to code. The guidelines instruct such passages to be treated as not instantiated. Applying this rule at the dataset level rather than per rater removes the need to arbitrate between any two raters' idiosyncratic thresholds and retains only what converged under independent application of the instructions.

Crucially, the conservative filter is an analysis-time operation. The canonical `scenes.csv` and `codes.csv` retain every scene and every code with full provenance, and analyses at more liberal thresholds (e.g. treating single-rater codings as exploratory observations) can be re-run against the same dataset without any re-coding.

2.6 Diagnostic metrics

Inter-rater consistency is reported at two tiers:

Scene-individuation agreement — the proportion of canonical scenes whose `rater_status` is "both".

Conditional classification agreement — given `rater_status == "both"`, the proportion of `agreement_flags` rows with `agreement == AGREE`, computed separately at taxonomy depth 0 (section), 1 (category), and 2 (leaf).

Rater-style asymmetries are summarised by per-coder tag-usage counts in `rater_style.csv`, which flag cases where individual raters systematically over- or under-use particular taxonomic tags across all their trips.

2.7 Reproducibility and open data

The full consistency pipeline is available in the project repository under `1.Recoding/`. All stages are reproducible from the original rater spreadsheets and trip-report files: running `scripts/01_extract_taxonomy.py` through `scripts/07_binning.py` in order regenerates every output file. The 40 per-trip adjudication YAMLs and their worksheets provide a full audit trail from each rater's original excerpt to its canonical scene assignment. A `RULES.md` document specifies the naming convention and the iterative rules under which new structural patterns are added. The original rater spreadsheets remain unchanged in their source directories (`Psilocybin_source_coding/` and `Brugmansia_source_coding/`).

3 Results

3.1 Corpus and coding overview

3.2 Inter-rater consistency: scene individuation

3.3 Inter-rater consistency: attribute classification

3.4 Sources of rater discrepancy

A systematic audit of the 40-trip recoded dataset (`1.Recoding/AUDIT_AND_MITIGATIONS.md`) identifies eight distinct sources of inter-rater discrepancy, ranked here by their contribution to the total 650 disagreement events in the dataset. Sources D1 and D3 jointly account for the majority of the signal; D2 is the principal residual source that cannot be resolved by dataset-level filtering alone.

D1. Individuation-threshold mismatch. The single dominant driver of individuation-level disagreement. Raters apply systematically different thresholds to the question of whether a narrative passage constitutes a discrete hallucinatory scene. Solo-scene counts by coder range from 6 (Alessio) to 49 (Alessandra), a 8× spread. The asymmetry is concentrated in the Psilocybin 01–10 pair, where Alessandra individuated 41 scenes her partner Brendan did not (a 20.5× within-pair asymmetry). Content systematically included only by the permissive raters — brief perceptual amplifications ("everything glassy", "voices feel alien"), per-object fragments within already-shared scenes, metaphorical visionary content, and ambient colour changes — falls into the Case-2 category of the project guidelines (`Guidelines.pdf §5`), which the permissive raters treated as codable and the conservative raters did not.

D2. Taxonomy ambiguity in "Type of visual alteration". The principal driver of attribute-level disagreement and the stubbornest source overall. The four leaves in this section (illusion, hallucination of an incrusted object, hallucination of a detached object, full-blown immersive hallucination) are defined cleanly in the abstract but their boundaries are not disambiguated for realistic edge cases: amplified colour on a real object, breathing walls, closed-eye visionary worlds, and gradual morphing of real objects into hallucinated ones can reasonably carry two or more tags. On shared scenes, 65% of taxonomy rows under this section are flagged `A_ONLY` or `B_ONLY` rather than `AGREE`.

D3. Stylistic tag over-use and under-use. Per-rater base-rate variation on specific taxonomy items drives many false-positive disagreement flags. Per-trip `Modal status` usage ranges from 0.1 per trip (Noah) to 5.9 per trip (Francesco), approximately 50×. Similar but smaller asymmetries appear for the `Hallucination of an incrusted object` leaf. These are not semantic disagreements about content but procedural-discipline differences about which tags to reflexively attach.

D4. Scope-layer mismatch (scene-level vs trip-level). The same passage is coded scene-level by one rater and trip-level by the other. Recurring instances: delusional beliefs ("laws of physics had changed"), ego-dissolution experiences with imagery ("cosmic winds blew through the centre of where I had been"), anthropomorphised objects ("the play structure flipped me off"), and insight-imagery ("the universe as a mycelium"). The PDF taxonomy places `Sense of reality`, `Insight`, `Ego dissolution`, `Own body distortion` and `Domain-specific violation` at trip-level, but the instructions do not explicitly direct raters how to handle passages that also have a scene-level perceptual component.

D5. Per-object excerpt fragmentation style. Alessandra's coding practice included listing each distinct object class within a scene as a separate scene-level row (e.g. "tree", "pine needles", "blue veins" as three excerpts within one scene). This produced 66 scene-level excerpts for `Psilocybe_02` alone versus 12 from Brendan covering the same content. This is a row-inflation style rather than a semantic disagreement; the PDF allows counting each object-class once per scene but does not prescribe a single-excerpt-per-item row structure.

D6. Lump-versus-split of evolving scenes. When a hallucinatory experience morphs over time (phase transitions within an NDE, repeated similar incidents, gradual evolution of a visionary world), some raters record one scene while others split the episode into phases. The PDF explicitly favours lumping — "a scene can change a little and still remains the same scene" — making splitting a minor rule deviation. In the current dataset only one case exhibited a clean lump–split pattern (`Brugmansia_12_S08`, retrofitted with the `LUMPA/SPLITB` convention); more latent cases likely exist but do not dominate the error budget.

D7. Somatic / interoceptive sensations coded as scenes. The PDF taxonomy places embodied phenomena (own-body distortion, arousal, ego-dissolution/bodily self) at trip-level. Susana systematically codes such passages ("de-animating into stone", "becoming a feather", "heartbeat irregularity") as scene-level hallucinations, while her pair partner Francesco keeps them at trip-level. The PDF does not provide a scene-level equivalent for embodied content, so this is partly a taxonomy gap rather than a rater error.

D8. Cognitive insights and revelations coded as scenes. `Insight` is an explicit trip-level taxonomy item, but when an insight arrives accompanied by imagery ("the connection I visualised was a growing mycelium"), raters split on whether to code the passage at scene-level as a visual hallucination or at trip-level as existential insight. This overlaps with D4 but is specifically about insight-content with imagistic delivery.

3.5 Mitigation: outcome of the conservative consensus filter

3.6 Phenomenological profile of psilocybin

3.7 Phenomenological profile of brugmansia

3.8 Direct comparison

4 Discussion

4.1 Phenomenological findings

4.2 Methodological contribution: a scene-ID recoding framework

We present a post-hoc framework that converts rater-level spreadsheet codings into a scene-identifier dataset with explicit status suffixes (`_AB`, `_A`, `_B`) and derived agreement tables. The

framework makes three tiers of inter-rater agreement separately measurable — did both raters see a hallucinatory scene in the passage, did they individuate it with the same boundaries, did they attach the same attributes — and supports both strict consensus and liberal analyses by filtering on the canonical data rather than modifying it. The convention is documented iteratively in `RULES.md` and is extensible to the other ten hallucinogens in the larger project without schema changes.

4.3 Sources of inter-rater discrepancy and mitigation

A systematic audit of the 40-trip dataset identifies the dominant sources of disagreement. Of the 650 total disagreement events, 19% are individuation-level (raters disagree on whether a passage is a hallucinatory scene) and 81% are attribute-level (raters agree a scene is present but tag it differently). This indicates that the bulk of the inter-rater problem is not failure to identify hallucinatory content but variation in how that content is labelled against the taxonomy.

Within individuation disagreement, the single largest driver is a *rater-threshold mismatch*: some raters (Alessandra most notably) treat brief perceptual amplifications, embodied sensations, and insight-imagery as individuated scenes, while others require a discrete object-level hallucinatory event. This asymmetry is concentrated in one coder pair (Psilocybin 01–10, Alessandra × Brendan), where Alessandra individuated 41 scenes her partner did not. The project's own guidelines (`Guidelines.pdf §5`) mandate a conservative Case-2 rule — if a passage is not clearly instantiated, leave it uncoded — and a critical reading of the deviations is that Alessandra's threshold is below the guideline's intended level. Classification-level disagreement is driven principally by ambiguity in the "Type of visual alteration" section (illusion, incrusted, detached, immersive), stylistic over-use or under-use of "Modal status" and "Incrusted object" tags, and a scope-layer mismatch in which the same passage is coded scene-level by one rater and trip-level by the other.

We assessed eight candidate mitigation strategies (conservative consensus filter, rater reconciliation, LLM-assisted adjudication, weighted κ, third-rater tiebreaker, scope-layer normalisation, taxonomy collapse, per-rater calibration; detailed in supplementary materials). Our recommendation is a tiered application: mechanical pre-processing (scope-layer normalisation + within-rater object-row deduplication) at consolidation time; a conservative consensus filter (`rater_status=both AND agreement=AGREE`) as the primary analysis base; a parallel liberal analysis reporting single-rater codings as exploratory observations with provenance; and, where feasible, a targeted reconciliation pass on the Alessandra × Brendan pair following the original project guidelines' prescribed method. Our view is that the framework reported here is scaffolding, not a substitute, for the reconciliation step the original guidelines foresaw.

4.4 Limitations

4.5 Implications and future directions

References